Signals and systems:

$H\{\{x(t)\}_{t\in\mathbb{R}}\} = \{y(t)\}_{t\in\mathbb{R}}$

$H\{x(t)\} = y(t)$

- Continuous-time: $t \in \mathbb{R}$

$\{x(t)\}_{t\in\mathbb{R}} \rightarrow \boxed{\mathcal{H}_\mathsf{c}} \rightarrow \{y(t)\}_{t\in\mathbb{R}}$

signals: $\{x(t)\}_{t\in\mathbb{R}}$, $\{y(t)\}_{t\in\mathbb{R}}$
system: $\mathcal{H}_\mathsf{c}$

- Discrete-time: $n \in \mathbb{Z}$

$\{x[n]\}_{n\in\mathbb{Z}} \rightarrow \boxed{\mathcal{H}} \rightarrow \{y[n]\}_{n\in\mathbb{Z}}$

signals: $\{x[n]\}_{n\in\mathbb{Z}}$, $\{y[n]\}_{n\in\mathbb{Z}}$
system: $\mathcal{H}$

- Assumption:

  All signals are *complex valued*. Below, $j \triangleq \sqrt{-1}$:
  $x[n] = \mathrm{Re}(x[n]) + j\,\mathrm{Im}(x[n]) = |x[n]|\, e^{j\angle x[n]}$.

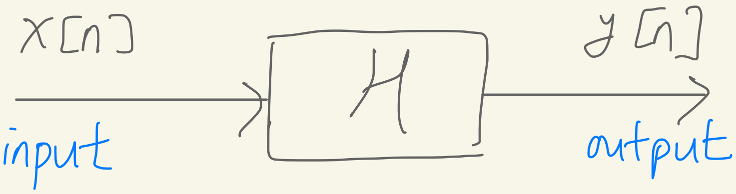
- Uniform sampling:

  We say that $\{x[n]\}_{n\in\mathbb{Z}}$ is a T-*sampled version* of $\{x(t)\}_{t\in\mathbb{R}}$ when $x[n] = x(nT)$ for every integer n. Here, T denotes the sampling interval in seconds.

- Key question:

  What "information" does the discrete-time sequence $\{x[n]\}_{n\in\mathbb{Z}}$ contain relative to the continuous-time waveform $\{x(t)\}_{t\in\mathbb{R}}$?

Convention in signal processing



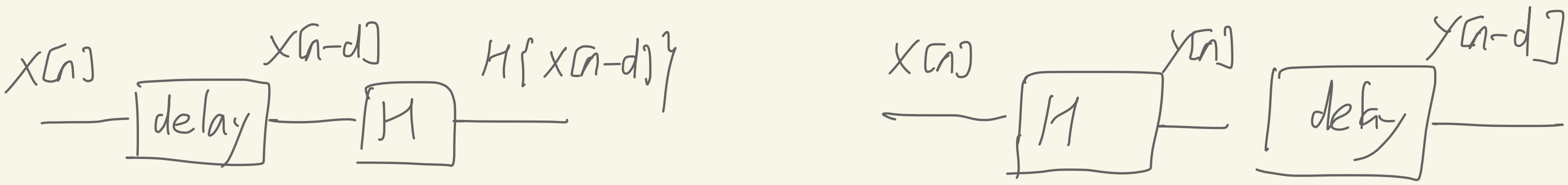
$$\{y[n]\} = H\{\{x[n]\}\}$$

Delay by d

$$H\{x[n]\} = x[n-d]$$

H is time invariant if

$$H\{x[n-d]\} = y[n-d]$$



for any d for any x[n]

A few important system properties:

For the statements below, assume that $\{y[n]\} = \mathcal{H}\{\{x[n]\}\}$.

1. Linear:
$$\mathcal{H}\{\underbrace{\alpha\{x[n]\} + \beta\{w[n]\}}_{\text{linear combination}}\} = \alpha\mathcal{H}\{\{x[n]\}\} + \beta\mathcal{H}\{\{w[n]\}\}$$
$$\forall \alpha, \beta \in \mathbb{C}, \ \forall\{x[n]\}, \{w[n]\}$$

2. Time-invariant:
$$\mathcal{H}\{\{x[n-d]\}\} = \{y[n-d]\}, \quad \forall d \in \mathbb{Z}, \ \forall\{x[n]\}$$

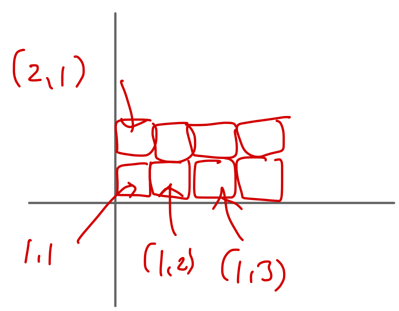
3. Causal:
At any time m, the output value $y[m]$ does not depend on the future input values $\{x[n]\}_{n>m}$.

4. BIBO Stable:
If $\{x[n]\}$ is bounded, then $\{y[n]\}$ will be bounded.

Recall that "bounded $\{x[n]\}$" means that there exists some finite value M_x such that $|x[n]| < M_x$ for all n.

2D signal

$x[m,n]$



index

(1,1) (1,2) (2,1)

total ordering

$\forall m,n \in \mathbb{Z}$ we can compare m and n so that $m=n$, $m>n$, $m<n$

Note: These properties can be directly re-stated for continuous-time systems as well.

Example of system properties:

Consider the system $y(t) = |x(t)|^2$:

- Is it linear?

  Linear combining at input: $|\alpha x(t) + \beta w(t)|^2$

  Linear combining at output: $\alpha |x(t)|^2 + \beta |w(t)|^2$

  The two are not equivalent, thus system not linear!

- Is it time-invariant?

  Delay at input: $|\text{delay}_\tau\{x(t)\}|^2 = \{|x(t-\tau)|^2\}$

  Delay at output: $\text{delay}_\tau\{|x(t)|^2\} = \{|x(t-\tau)|^2\}$

  The two are equivalent, thus system is time-invariant.

- Is it causal?

  For all t, does $y(t)$ depend only on $\{x(\tau)\}_{\tau \leq t}$?

  Yes, thus system is causal.

- Is it stable?

  Do all bounded inputs yield bounded outputs?

  If $|x(t)| < M_x$ for all $t \in \mathbb{R}$, is there an M_y such that $|y(t)| < M_y$ for all $t \in \mathbb{R}$?

  Yes: $M_y = M_x^2$ suffices. Thus system is stable.

More system examples:

1. Modulation: $y[n] = e^{j\omega_0 n} x[n]$ $\quad Y(e^{j\omega}) = X(e^{j(\omega-\omega_0)})$

   linear? time-invariant? causal? stable?

   yes, no, yes, yes $\quad \dfrac{Y(e^{j\omega})}{X(e^{j\omega})} = \dfrac{X(e^{j(\omega-\omega_0)})}{X(e^{j\omega})}$

2. Convolution: $y(t) = \int_{-\infty}^{\infty} h(\tau)x(t-\tau)d\tau$

   linear? time-invariant? causal? stable?

   yes, yes, depends on $h(t)$, depends on $h(t)$

   auto-regressive $\quad Y(e^{j\omega}) = X(e^{j\omega}) + a e^{-j\omega} Y(e^{j\omega})$

3. Recursion: $y[n] = x[n] + a\, y[n-1]$ $\quad \dfrac{Y(e^{j\omega})}{X(e^{j\omega})} = \dfrac{1}{1-ae^{-j\omega}}$

   linear? time-invariant? causal? stable?

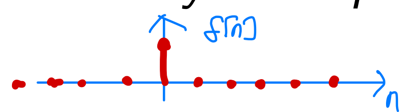
$$\begin{aligned} y[n] &= x[n] + a\big(x[n-1] + ay[n-2]\big) \\ &= x[n] + ax[n-1] + a^2\big(x[n-2] + ay[n-3]\big) \\ &= \textstyle\sum_{m=0}^{\infty} a^m x[n-m] \end{aligned}$$

yes, yes, yes, for $|a| < 1$, assuming finite $|y[-\infty]|$

LTI systems — input/output behavior:

- Linear time-invariant (LTI) systems are a very important and convenient class of systems:

  The input/output behavior of an LTI system is completely characterized by its impulse response!

- Discrete-time case: 

  $\delta[n] = \begin{cases} 1 & n=0 \\ 0 & n \neq 0 \end{cases}$

  – The *impulse response* of $\mathcal{H}$ is the signal $\{h[n]\} = \mathcal{H}\{\{\delta[n]\}\}$, for Kronecker delta $\{\delta[n]\}$.

  – The *Kronecker delta* sequence $\{\delta[n]\}$ is defined as

  $$\delta[n] \triangleq \begin{cases} 1 & \text{if } n = 0 \\ 0 & \text{if } n \neq 0 \end{cases},$$

  and obeys the following *sifting property*:

  $$y[n] = \sum_{m=-\infty}^{\infty} y[m]\delta[m-n] = \sum_{m=-\infty}^{\infty} y[m]\delta[n-m].$$

  – Why does the impulse response $\{h[n]\}$ completely characterize $\mathcal{H}$?

  First, write $\{x[n]\} = \sum_m x[m]\{\delta[n-m]\}$. Then

  $$\begin{aligned} \mathcal{H}\{\{x[n]\}\} &= \mathcal{H}\{\textstyle\sum_m x[m]\{\delta[n-m]\}\} \\ &= \textstyle\sum_m x[m]\mathcal{H}\{\{\delta[n-m]\}\} \quad \text{(linearity)} \\ &= \textstyle\sum_m x[m]\{h[n-m]\} \quad \text{(time-invariance)}. \end{aligned}$$

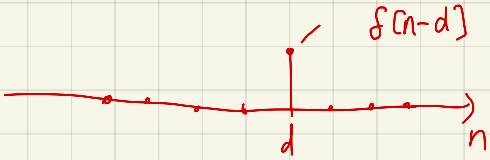
  – Thus, the system's output equals the *convolution* of the input $\{x[n]\}$ with the impulse response $\{h[n]\}$!

$$y[\ell] = \{y[n]\}\Big|_{n=\ell} = \sum_m x[m]\,\{h[n-m]\}\Big|_{n=\ell} = \sum_m x[m]\, h[\ell-m]$$

$$\{\delta[n]\} \cdot \{x[n]\} = \sum_{n=-\infty}^{\infty} \delta^*[n]\, x[n]$$

$$= \delta[0]\, x[0] = x[0]$$

$$\underbrace{\{\delta[n-d]\}}_{\text{shifted impulse}} \cdot \{x[n]\} = \underbrace{\sum_{n=-\infty}^{\infty} \delta^*[n-d]\, x[n] = x[d]}_{\text{sifting property of impulse.}}$$



$$x[d] = \sum_{n=-\infty}^{\infty} \delta[n-d]\, x[n]$$

$$= \sum_{m=-\infty}^{\infty} \delta[m-d]\, x[m] \qquad \text{true for all } d.$$

$$\underbrace{x[n]}_{\text{evaluation of signal } x \text{ at time } n} = \sum_{m=-\infty}^{\infty} \delta[m-n]\, x[m] \qquad \text{for all } n$$

$$\vdots$$

$$x[-1] = \sum_m \delta[m+1]\, x[m]$$

$$x[0] = \sum_m \delta[m-0]\, x[m]$$

$$x[1] = \sum_m \delta[m-1]\, x[m]$$

$$\vdots$$

$$\{x[n]\} = \sum_{m=-\infty}^{\infty} \underbrace{\{\delta[m-n]\}}_{\text{function of } n}\, x[m]$$

linear algebra analog

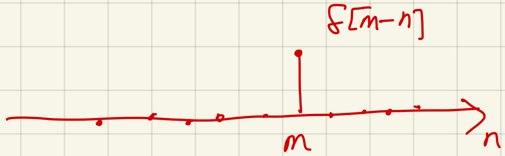
$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

$$= x_1 \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} + x_3 \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

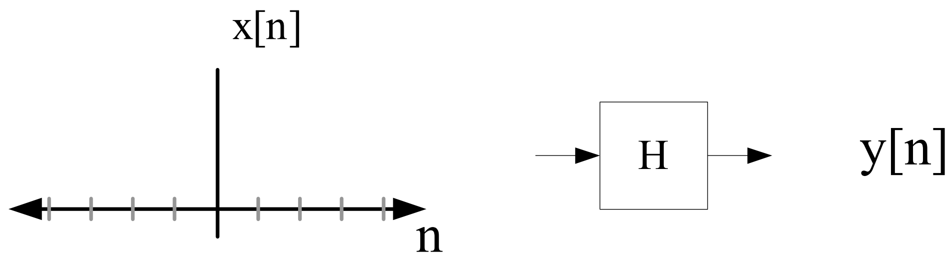
synthesizing $\vec{x}$ as a linear combination of

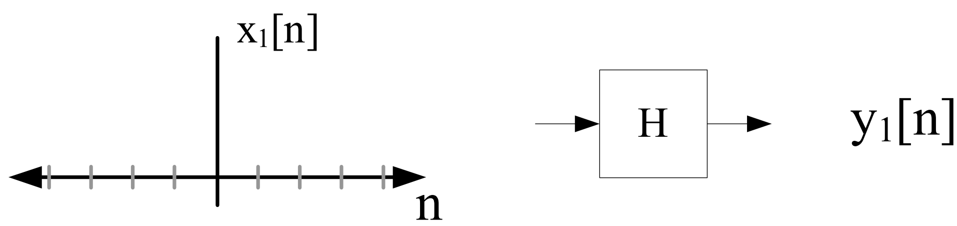
$$\begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

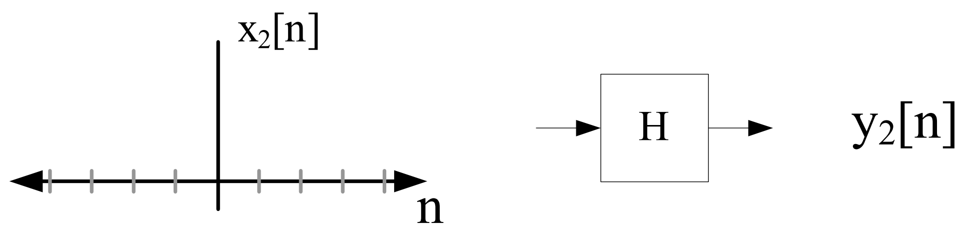
$\{\delta[m-n]\}$

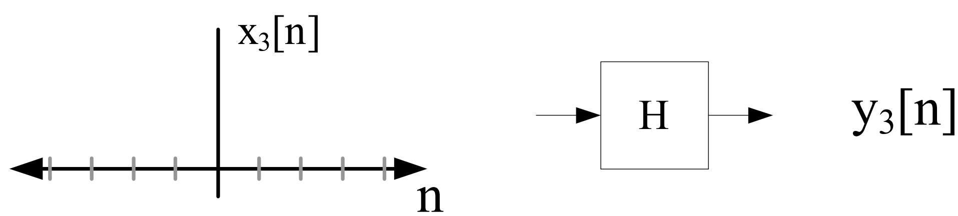


Example:





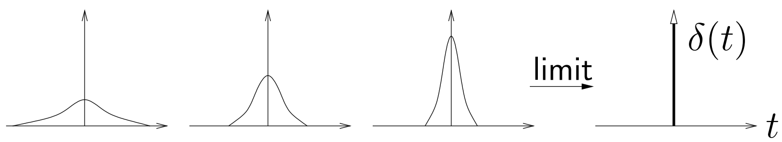




$\cancel{2\delta(t) = \delta(t)}$?

$\cancel{\delta(t) = \begin{cases} \infty & t=0 \\ 0 & t \neq 0 \end{cases}}$

- Continuous-time case:
  - The *impulse response* of $\mathcal{H}_\mathsf{c}$ is the signal $\{h(t)\} = \mathcal{H}_\mathsf{c}\{\{\delta(t)\}\}$, for Dirac delta $\{\delta(t)\}$.
  - The *Dirac delta* waveform can be understood as the limit of increasingly skinny *unit-area* pulses:



    and it obeys the following *sifting property*:

$$y(t) = \int_{-\infty}^{\infty} y(\tau)\delta(\tau - t)d\tau = \int_{-\infty}^{\infty} y(\tau)\delta(t-\tau)d\tau.$$

  - Why does the impulse response $\{h(t)\}$ completely characterize $\mathcal{H}_\mathsf{c}$?
    First, write $\{x(t)\} = \int x(\tau)\{\delta(t-\tau)\}d\tau$. Then

$$\begin{aligned} \mathcal{H}_\mathsf{c}\{\{x(t)\}\} &= \mathcal{H}_\mathsf{c}\{\textstyle\int x(\tau)\{\delta(t-\tau)\}d\tau\} \\ &= \textstyle\int x(\tau)\mathcal{H}_\mathsf{c}\{\{\delta(t-\tau)\}\}d\tau \quad \text{(linearity)} \\ &= \textstyle\int x(\tau)\{h(t-\tau)\}d\tau \quad \text{(time-invariance).} \end{aligned}$$

  - Thus, the system's output equals the *convolution* of the input $\{x(t)\}$ with the impulse response $\{h(t)\}$!

- We usually denote the convolution operator by "$*$":

$$\begin{aligned} \textstyle\int x(\tau)\{h(t-\tau)\}d\tau &= \{x(t)\} * \{h(t)\} \\ \textstyle\sum_m x[m]\{h[n-m]\} &= \{x[n]\} * \{h[n]\} \end{aligned}$$

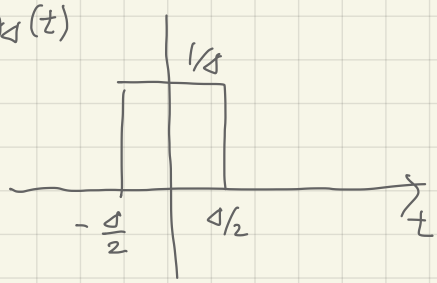
$* : V \times V \longrightarrow V$

binary operator

$$\{\delta(t)\}\cdot\{x(t)\} = \int_{-\infty}^{\infty} \delta(t)\,x(t)\,dt = x(0)$$

$$\{\delta[n]\}\cdot\{x[n]\} = \sum_{n=-\infty}^{\infty} \delta[n]\,x[n] = x[0]$$

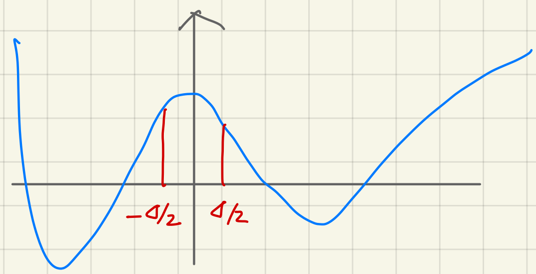
$\delta_\Delta(t)$



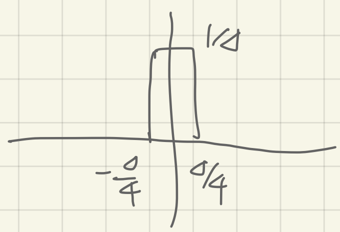
$$\int_{-\infty}^{\infty} \delta_\Delta(t)\,dt = 1$$

$$\boxed{\delta(t) = \lim_{\Delta\to 0} \delta_\Delta(t)} \quad ?$$

$$\{\delta_\Delta(t)\}\cdot\{x(t)\} = \underbrace{\frac{1}{\Delta}\int_{-\Delta/2}^{\Delta/2} x(t)\,dt}_{\text{average of } x(t) \text{ on the interval } [-\frac{\Delta}{2},\, \frac{\Delta}{2}]}$$



$\delta_\Delta(2t)$



$$\int_{-\infty}^{\infty} \delta_\Delta(2t)\,dt = \frac{1}{2}$$

$$\lim_{\Delta\to 0} \delta_\Delta(2t) = \frac{1}{2}\,\delta(t)$$

$$\parallel$$

$$\delta(2t)$$

$$\{\delta(t)\} \cdot \{x(t)\} = \int_{-\infty}^{\infty} \delta(t)\, x(t)\, dt = x(0)$$

$$\{\delta'(t)\} \cdot \{x(t)\} = \int_{-\infty}^{\infty} \delta'(t)\, x(t)\, dt$$

$$= \underbrace{\cancelto{0}{\delta(t)\, x(t) \Big|_{t=-\infty}^{\infty}}} - \underbrace{\int_{-\infty}^{\infty} \delta(t)\, x'(t)\, dt}_{x'(0)}$$

$$\int_a^b f'g = fg\Big|_a^b - \int_a^b fg'$$

$$\{\delta'(t)\} \cdot \{x(t)\} = -x'(0)$$

$$\{\delta''(t)\} \cdot \{x(t)\} = x''(0)$$

Computing Convolution:

1. Let the computer do it.

In Matlab,

```
>> y = conv(h,x);
```

Example:

$x[n] = u[n] - u[n-5]$
$h[n] = 2\delta[n] - \delta[n-1] + 3\delta[n-2]$

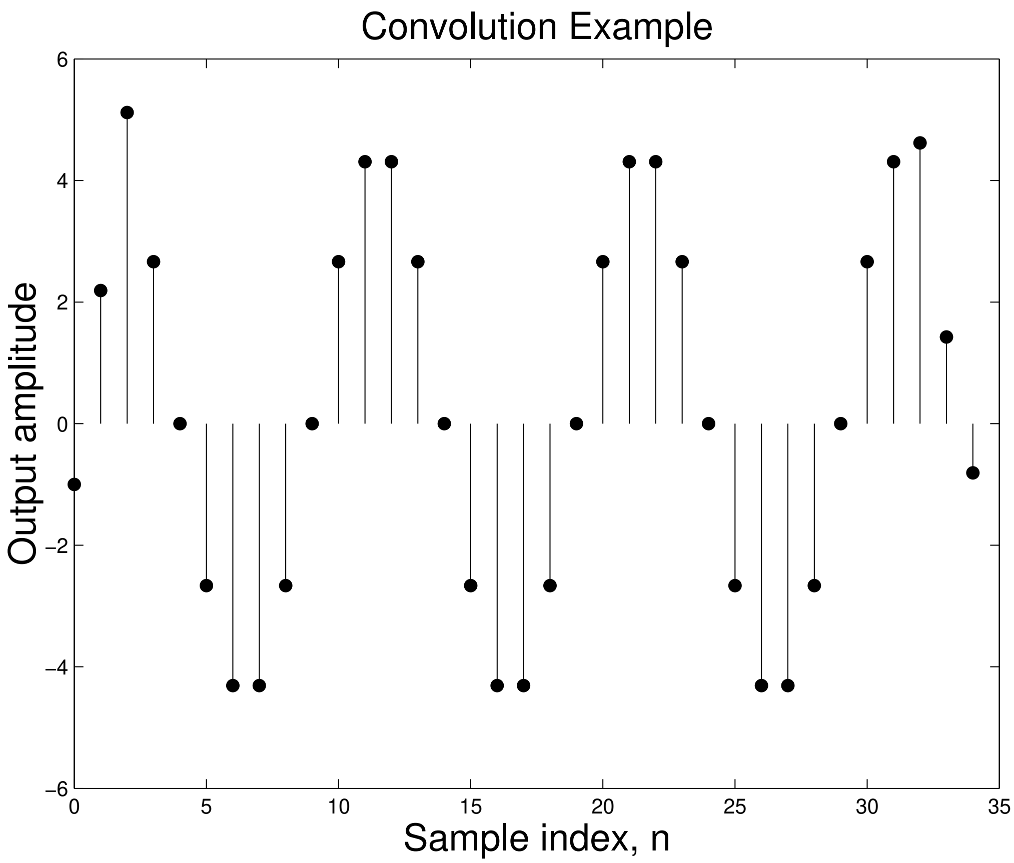
```
>> x = [1, 1, 1, 1, 1];
>> h = [2, -1, 3];
>> y = conv(x,h)

y =

     2     1     4     4     4     2     3
```

Matlab Example:

```
>> n = [0:1:31];
>> x = cos(0.2*pi*n);% cosine, amplitude 1, 0.2*pi radians
>> h = [-1 3 3 -1 ];% impulse response
>> y = conv(x,h)
>> % construct a plot
>> stem([0:length(y)-1], y,'filled');
```



Computing Convolution:

2. Graphical convolution.

"Flip, shift, multiply, add."

Very useful for a fast, qualitative, visual interpretation.

Discrete convolution demo in Matlab: http://dspfirst.gatech.edu/matlab/

Computing Convolution:

3. Matrix-vector multiply (for finite duration signals)

$$y[n] = \sum_k x[k]\, h[n-k] = \sum_k h[k]\, x[n-k]$$

Example: $M = 3$; $x[n]$ zero for $n < 0$ and $n > 7$.

Toeplitz matrix.

$$\begin{bmatrix} x[0] & 0 & 0 & 0 \\ x[1] & x[0] & 0 & 0 \\ x[2] & x[1] & x[0] & 0 \\ x[3] & x[2] & x[1] & x[0] \\ x[4] & x[3] & x[2] & x[1] \\ x[5] & x[4] & x[3] & x[2] \\ x[6] & x[5] & x[4] & x[3] \\ x[7] & x[6] & x[5] & x[4] \\ 0 & x[7] & x[6] & x[5] \\ 0 & 0 & x[7] & x[6] \\ 0 & 0 & 0 & x[7] \end{bmatrix} \begin{bmatrix} h[0] \\ h[1] \\ h[2] \\ h[3] \end{bmatrix} = \begin{bmatrix} y[0] \\ y[1] \\ y[2] \\ y[3] \\ y[4] \\ y[5] \\ y[6] \\ y[7] \\ y[8] \\ y[9] \\ y[10] \end{bmatrix}$$